

Table 1: Sequential MAS results for Qwen3-8B. Acc. is accuracy (%), Token values for Single, TextMAS, and Linear are taken from the reference figures. Time for $m \in \{\text{Single, TextMAS, Linear}\}$ is converted to the H200 scale by $T_m^{(H)} = N_m \rho_H$. We use the same-dataset Kernel time-per-token ratio when available and otherwise the H200 Identical ratio. Entries without either H200 ratio are denoted by $-$. Best results in each row are bold. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by $-$.

Tasks	Metrics	Baselines		LatentMAS variants					Improve
		Single	TextMAS	Identical	Linear	Kernel	Kernel-ES	Soft	
Sequential MAS Setting									
AIME24	Acc.	50.0	53.3	—	56.7	66.25	64.17	63.75	↑ 13.0
	Token	12891	38596	—	8953	8548	8770	22342	↓ 77.9%
	Time	340.2	1018.5	—	236.3	225.6	225.8	549.0	× 4.5
AIME25	Acc.	46.7	53.3	—	53.3	49.58	50.42	49.17	↓ 2.9
	Token	14692	45088	—	8699	10158	9830	24557	↓ 78.2%
	Time	377.8	1159.5	—	223.7	261.2	255.5	597.0	× 4.5
ARC-C	Acc.	91.0	94.6	94.54	94.4	94.80	95.56	—	↑ 1.0
	Token	846	2252	559	529	570	915	—	↓ 74.7%
	Time	5.5	14.6	3.7	3.4	3.7	4.5	—	× 3.9
ARC-E	Acc.	95.6	99.1	98.27	98.8	98.32	98.40	—	↓ 0.7
	Token	656	2085	468	490	472	822	—	↓ 77.4%
	Time	3.5	11.1	2.5	2.6	2.5	3.2	—	× 4.4
GPQA	Acc.	39.9	43.4	—	45.5	48.74	48.61	—	↑ 5.3
	Token	6435	17986	—	4571	3588	3846	—	↓ 80.1%
	Time	104.5	292.0	—	74.2	58.3	59.4	—	× 5.0
GSM8K	Acc.	81.1	92.3	91.58	93.8	91.74	92.19	—	↓ 0.1
	Token	1280	2324	628	860	642	1029	—	↓ 72.4%
	Time	8.2	15.0	4.1	5.5	4.1	5.2	—	× 3.7
HumanEval+	Acc.	74.4	80.5	84.15	80.5	83.54	83.38	82.62	↑ 3.0
	Token	2507	4593	1668	1866	1637	1696	15055	↓ 64.4%
	Time	28.9	52.9	18.9	21.5	18.8	21.5	112.4	× 2.8
MBPP+	Acc.	64.8	69.5	—	74.6	75.26	75.20	73.15	↑ 5.8
	Token	2053	3695	—	1164	1462	1528	14388	↓ 60.4%
	Time	25.2	45.3	—	14.3	17.9	19.6	101.6	× 2.5
MedQA	Acc.	53.0	75.0	76.33	75.3	73.67	72.00	66.67	↓ 1.3
	Token	2098	4260	1872	1555	1139	1154	15972	↓ 73.3%
	Time	23.0	46.7	17.0	17.0	12.5	13.6	111.1	× 3.7

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.

Table 2: Hierarchical MAS results for Qwen3-8B. Acc. is accuracy (%), Token values for Single, TextMAS, and Linear are taken from the reference figures. Time for $m \in \{\text{Single, TextMAS, Linear}\}$ is converted to the H200 scale by $T_m^{(H)} = N_m \rho_H$. We use the same-dataset Kernel time-per-token ratio when available and otherwise the H200 Identical ratio. Entries without either H200 ratio are denoted by $-$. Best results in each row are bold. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by $-$.

Tasks	Metrics	Baselines		LatentMAS variants					Improve
		Single	TextMAS	Identical	Linear	Kernel	Kernel-ES	Soft	
<i>Hierarchical MAS Setting</i>									
AIME24	Acc.	50.0	53.3	—	53.3	60.00	65.83	—	↑ 12.5
	Token	12891	42629	—	7526	9926	9898	—	↓ 76.8%
	Time	335.6	1109.8	—	195.9	258.4	257.9	—	× 4.3
AIME25	Acc.	46.7	50.0	—	50.0	53.33	52.50	—	↑ 3.3
	Token	14692	53929	—	13230	11207	11300	—	↓ 79.2%
	Time	378.9	1390.9	—	341.2	289.1	291.9	—	× 4.8
ARC-C	Acc.	91.0	93.3	94.20	93.9	95.05	95.05	—	↑ 1.8
	Token	846	2854	606	344	603	822	—	↓ 78.9%
	Time	5.3	17.8	3.9	2.1	3.8	4.3	—	× 4.7
ARC-E	Acc.	95.6	98.2	98.48	98.3	98.23	98.23	—	↑ 0.0
	Token	656	2237	494	308	495	721	—	↓ 77.9%
	Time	3.4	11.5	2.6	1.6	2.5	3.0	—	× 4.6
GPQA	Acc.	39.9	43.0	—	46.9	51.39	49.75	—	↑ 8.4
	Token	6435	22450	—	3395	3752	3964	—	↓ 83.3%
	Time	102.8	358.5	—	54.2	59.9	61.5	—	× 6.0
GSM8K	Acc.	81.1	90.4	91.58	89.5	91.28	92.34	—	↑ 1.9
	Token	1280	2370	688	353	704	932	—	↓ 70.3%
	Time	7.9	14.7	4.1	2.2	4.4	4.8	—	× 3.3
HumanEval+	Acc.	74.4	76.8	82.62	78.0	83.84	83.23	—	↑ 7.0
	Token	2507	8768	1688	1274	1681	1637	—	↓ 81.3%
	Time	30.0	104.8	20.2	15.2	20.1	21.0	—	× 5.2
MBPP+	Acc.	64.8	71.9	70.97	72.2	72.75	72.16	73.21	↑ 0.8
	Token	2053	7703	1436	1264	1502	1460	12768	↓ 81.0%
	Time	27.5	103.1	20.3	16.9	20.1	20.2	92.2	× 5.1
MedQA	Acc.	53.0	76.3	75.33	77.0	73.33	72.67	—	↓ 3.0
	Token	2098	6893	1391	1007	1351	1336	—	↓ 80.6%
	Time	24.4	80.2	15.5	11.7	15.7	14.8	—	× 5.4

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.

Table 3: Sequential MAS results for Qwen3-14B. Acc. is accuracy (%), Token values for Single, TextMAS, and Linear are taken from the reference figures. Time for $m \in \{\text{Single, TextMAS, Linear}\}$ is converted to the H200 scale by $T_m^{(H)} = N_m \rho_H$. We use the same-dataset Kernel time-per-token ratio when available and otherwise the H200 Identical ratio. Entries without either H200 ratio are denoted by $-$. Best results in each row are bold. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by $-$.

Tasks	Metrics	Baselines		LatentMAS variants					Improve
		Single	TextMAS	Identical	Linear	Kernel	Kernel-ES	Soft	
Sequential MAS Setting									
AIME24	Acc.	63.3	63.3	—	66.7	72.08	69.58	—	↑ 8.8
	Token	11263	32092	—	10593	8259	8489	—	↓ 74.3%
	Time	348.6	993.3	—	327.9	255.6	283.4	—	× 3.9
AIME25	Acc.	56.7	60.0	—	63.3	56.67	60.00	49.58	↑ 0.0
	Token	11298	44618	—	11402	10266	10402	21403	↓ 77.0%
	Time	369.3	1458.3	—	372.7	335.5	350.6	575.5	× 4.3
ARC-C	Acc.	92.6	95.9	95.99	95.6	95.48	95.73	—	↓ 0.2
	Token	773	2985	463	426	461	544	—	↓ 84.6%
	Time	7.2	27.7	4.3	4.0	4.3	6.0	—	× 6.5
ARC-E	Acc.	97.2	99.0	98.48	99.4	98.82	98.57	—	↓ 0.2
	Token	608	1670	379	224	380	462	—	↓ 77.3%
	Time	4.3	11.8	2.8	1.6	2.7	4.5	—	× 4.4
GPQA	Acc.	48.5	51.5	—	52.0	—	—	—	—
	Token	5547	12676	—	5454	—	—	—	—
	Time	—	—	—	—	—	—	—	—
GSM8K	Acc.	83.7	93.8	92.12	95.2	—	—	—	—
	Token	1118	3324	577	644	—	—	—	—
	Time	9.8	29.0	5.0	5.6	—	—	—	—
HumanEval+	Acc.	76.8	81.1	86.89	86.5	—	—	—	—
	Token	2366	5934	1590	2042	—	—	—	—
	Time	35.9	90.0	24.1	31.0	—	—	—	—
MBPP+	Acc.	68.5	72.8	78.17	75.7	—	—	—	—
	Token	1858	4971	1345	1621	—	—	—	—
	Time	30.2	80.8	21.9	26.3	—	—	—	—
MedQA	Acc.	64.7	80.3	81.00	80.7	—	—	—	—
	Token	1746	3444	1616	1841	—	—	—	—
	Time	20.6	40.6	19.0	21.7	—	—	—	—

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.

Table 4: Hierarchical MAS results for Qwen3-14B. Acc. is accuracy (%), Token values for Single, TextMAS, and Linear are taken from the reference figures. Time for $m \in \{\text{Single, TextMAS, Linear}\}$ is converted to the H200 scale by $T_m^{(H)} = N_m \rho_H$. We use the same-dataset Kernel time-per-token ratio when available and otherwise the H200 Identical ratio. Entries without either H200 ratio are denoted by $-$. Best results in each row are bold. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by $-$.

Tasks	Metrics	Baselines		LatentMAS variants					Improve
		Single	TextMAS	Identical	Linear	Kernel	Kernel-ES	Soft	
<i>Hierarchical MAS Setting</i>									
AIME24	Acc.	63.3	70.0	—	73.3	69.17	69.58	71.25	↓ 0.4
	Token	11263	29025	—	10230	8947	9017	23924	↓ 69.2%
	Time	357.6	921.7	—	324.8	284.1	291.0	639.9	× 3.2
AIME25	Acc.	56.7	66.7	—	66.7	59.17	57.92	—	↓ 7.5
	Token	11298	50003	—	9527	10755	11122	—	↓ 78.5%
	Time	364.4	1612.8	—	307.3	346.9	365.7	—	× 4.6
ARC-C	Acc.	92.6	95.3	95.65	95.5	95.99	95.65	—	↑ 0.7
	Token	773	2167	434	295	424	477	—	↓ 80.4%
	Time	6.7	18.8	4.0	2.6	3.7	5.4	—	× 5.1
ARC-E	Acc.	97.2	98.3	98.70	98.7	98.78	98.23	—	↑ 0.5
	Token	608	2752	344	619	337	402	—	↓ 87.8%
	Time	4.4	19.7	2.5	4.4	2.4	4.0	—	× 8.2
GPQA	Acc.	48.5	52.0	—	53.0	—	—	—	—
	Token	5547	20931	—	3606	—	—	—	—
	Time	—	—	—	—	—	—	—	—
GSM8K	Acc.	83.7	90.8	91.96	91.6	—	—	—	—
	Token	1118	3021	574	495	—	—	—	—
	Time	9.0	24.4	4.6	4.0	—	—	—	—
HumanEval+	Acc.	76.8	84.1	88.41	86.6	—	—	—	—
	Token	2366	8114	1473	1512	—	—	—	—
	Time	38.2	131.1	23.8	24.4	—	—	—	—
MBPP+	Acc.	68.5	73.0	76.92	73.8	—	—	—	—
	Token	1858	7458	1209	1187	—	—	—	—
	Time	35.8	143.7	23.3	22.9	—	—	—	—
MedQA	Acc.	64.7	78.0	81.33	78.3	—	—	—	—
	Token	1746	5473	870	899	—	—	—	—
	Time	24.3	76.3	12.1	12.5	—	—	—	—

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.